
Smart Load Vision System

AI-Powered Loading Optimization

Phase I : Proof of Concept



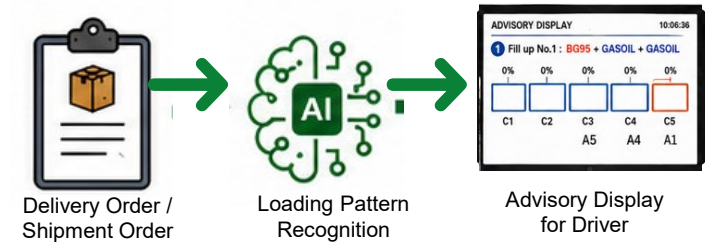
Objective

1. Overview of Smart Load Vision System project
2. Site survey at gantry #15 & Server readiness
3. Discussion on installation feasibility, including camera/HMI location, power, LAN, and hazardous area requirements
4. Alignment on safety requirements and next steps

Overview of Smart Load Vision System

AI advisory + Computer vision detection to **control** loading pattern and **reduce loading time** variation

1. ADVISORY



AI recommends the optimal loading pattern based on order details, compartment configuration, and gantry constraints.

- ✓ Recommend **Optimal loading pattern**
- ✓ **Notify** when actual behavior **deviates** from the **optimal loading pattern**

2. DRIVER EXECUTION

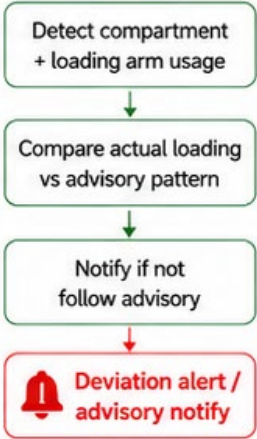


- Check advisory display first
- Connect Loading arm to Correct compartment
- Load as recommend pattern

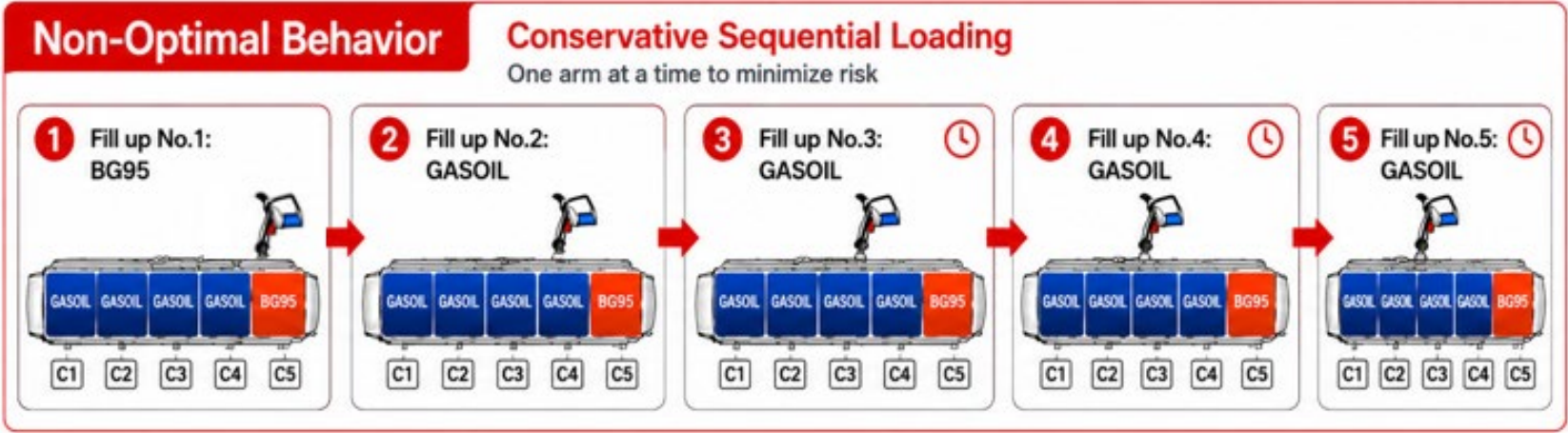
3. DETECTION & MONITORING



- Detect physical **compartments** and active **Loading arms**
- Check whether driver **follows** the **recommended pattern**

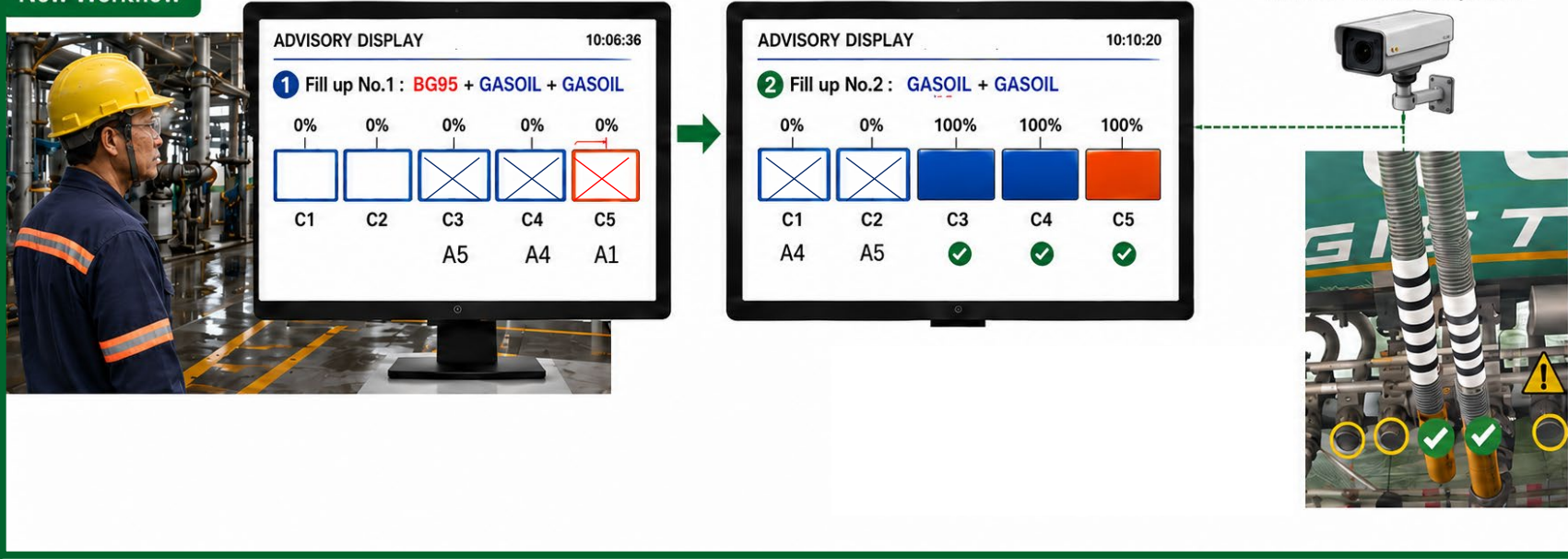


Overview of Smart Load Vision System



TO-BE / New Workflow

AI-powered advisory, optimized loading sequence



AI camera checks actual loading arm vs recommended pattern

A4

A5

✓

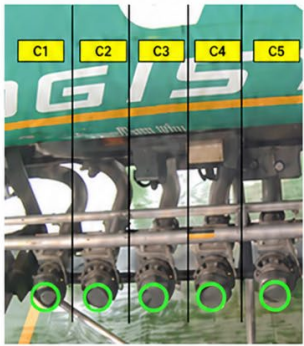
✓

✓

How Computer Vision Works?

STEP 1

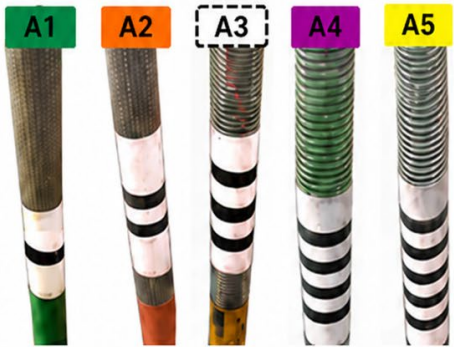
Detect Compartment (C1 - C5)



Detect
Compartment C1 - C5

STEP 2

Detect Arm Number (A1 - A5)



Detect arm number
(A1 - A5) based on
the number of black
stripes

STEP 3

Map Compartment + Arm Number as Recommend Pattern

	Fill up No.1			Fill up No.2	
Compartment	C5	C4	C3	C2	C1
Recommended Arm No.	A1	A4	A5	A5	A4
Actual Arm No.	N/A	A4	A5		
Pattern Check	-	✓	✓		

PoC project scope



PHASE I: Proof of Concept



Aug 2026 – Aug 2027 (12 month)



Gantry #15 (Infrastructure-ready site for fast and low-risk PoC execution)



Scope

- Install **Camera, Advisory display** and Computing setup (using existing ENIM's server)
- Develop **AI detection model** (Arm + Compartment)
- Generate **Advisory loading pattern** from Customer PO/TAS database
- Verify **driver execution** against recommended pattern



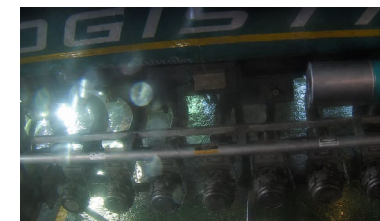
Key Deliverables

- PoC-ready **AI detection model** for loading arm and compartment
- **Advisory display logic** for recommended loading sequence
- Driver compliance rate monitoring

Key Focus :

Validate model reliability under real terminal conditions :

1. Various environment : Day/night, Glare, Rain, Shadows, Blur
2. Truck variation



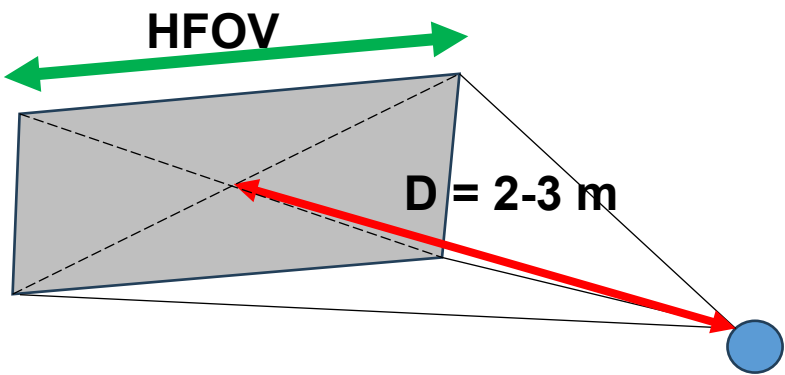
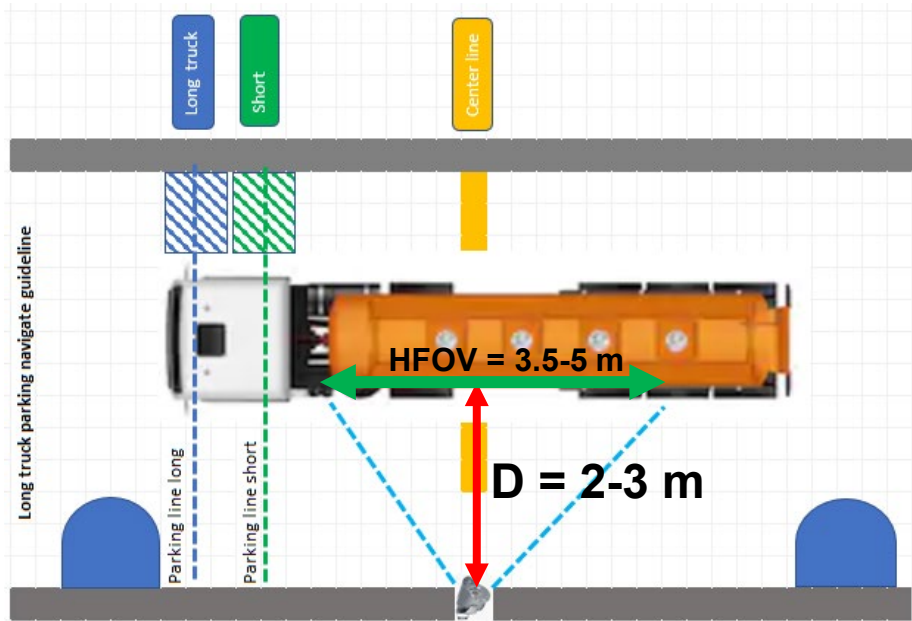
Example : Normalized Patterns (TOP30)

Config Format	Recommend Pattern	Total Count	NLL Count	LLTE Count
GASOIL(3)	GASOIL, GASOIL GASOIL	33245	10360	22885
GASOIL(5)	GASOIL, GASOIL GASOIL, GASOIL GASOIL	27586	10375	17211
GASOIL(4)	GASOIL, GASOIL GASOIL, GASOIL	22755	8667	14088
BG91, BG95, GASOIL(3)	BG91, GASOIL, GASOIL BG95, GASOIL	19065	7019	12046
BG95, GASOIL(4)	BG95, GASOIL, GASOIL GASOIL, GASOIL	16981	6686	10295
BG91, BG95(2), GASOIL(2)	BG95, GASOIL, GASOIL BG91, BG95	10304	3633	6671
BG91(2), BG95, GASOIL(2)	BG91, GASOIL, GASOIL BG91, BG95	10236	3477	6759
GASOIL(8)	GASOIL, GASOIL GASOIL, GASOIL GASOIL, GASOIL GASOIL, GASOIL	7635	2193	5442
BG95(2), GASOIL(3)	BG95, GASOIL, GASOIL BG95, GASOIL	7568	2936	4632
BG91, BG95, GASOIL(6)	BG91, GASOIL, GASOIL BG95, GASOIL, GASOIL GASOIL, GASOIL	6350	2997	3353
BG91, BG95(2), GASOIL(5)	BG95, GASOIL, GASOIL BG91, GASOIL, GASOIL BG95, GASOIL	5571	2600	2971
BG95(2), GASOIL(6)	BG95, GASOIL, GASOIL BG95, GASOIL, GASOIL GASOIL, GASOIL	5254	1844	3410
BG91, GASOIL(4)	BG91, GASOIL, GASOIL GASOIL, GASOIL	4543	1787	2756
BG91(2), BG95(2), GASOIL	BG91, BG95, GASOIL BG91, BG95	4502	1556	2946
BG91, BG95, GASOIL(2)	BG91, GASOIL, GASOIL BG95	4225	2027	2198
UG91(4)	UG91 UG91 UG91 UG91	3826	3826	0
GASOIL(7)	GASOIL, GASOIL GASOIL, GASOIL GASOIL, GASOIL GASOIL	3508	1154	2354
BG91(2), BG95, GASOIL(5)	BG91, GASOIL, GASOIL BG91, GASOIL, GASOIL BG95, GASOIL	3385	1693	1692
BG95, GASOIL(7)	BG95, GASOIL, GASOIL GASOIL, GASOIL GASOIL, GASOIL GASOIL	3170	1081	2089
BG91(2), BG95(2), GASOIL(4)	BG91, GASOIL, GASOIL BG95, GASOIL, GASOIL BG91, BG95	2584	1278	1306
BG91(2), BG95(2), GASOIL(3)	BG91, GASOIL, GASOIL BG91, BG95, GASOIL BG95	2567	898	1669
GASOIL(2)	GASOIL, GASOIL	2561	424	2137
BG95, GASOIL(3)	BG95, GASOIL, GASOIL GASOIL	2432	1377	1055
FOA(4)	FOA FOA FOA FOA	2394	2394	0
BG91(2), GASOIL(3)	BG91, GASOIL, GASOIL BG91, GASOIL	2109	790	1319
BG91, BG95, GASOIL(5)	BG91, GASOIL, GASOIL BG95, GASOIL, GASOIL GASOIL	2027	719	1308
BG91(2), BG95, GASOIL(4)	BG91, GASOIL, GASOIL BG91, GASOIL, GASOIL BG95	2025	627	1398
BG91, GASOIL(3)	BG91, GASOIL, GASOIL GASOIL	2000	1335	665
FOA(5)	FOA FOA FOA FOA FOA	1993	1993	0
FOC(4)	FOC, FOC FOC, FOC	1902	1902	0

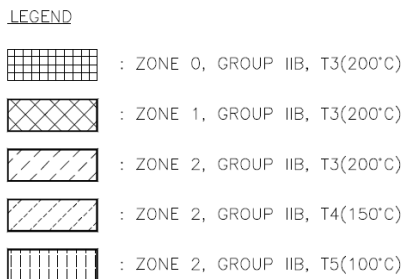
Total Config format = 373

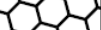


Site survey at gantry #15 & Server readiness



CUSTOMER
INSIGHTS



- | | |
|---|--------------------------------|
|  | : ZONE 0, GROUP IIB, T3(200°C) |
|  | : ZONE 1, GROUP IIB, T3(200°C) |
|  | : ZONE 2, GROUP IIB, T3(200°C) |
|  | : ZONE 2, GROUP IIB, T5(100°C) |

HMI Display

Required

Zone 1, Group IIB, T3 (200°C)



Existing HMI



Proposed new HMI

Item	Existing MERCURY2 HMI	BARTEC POLARIS PROFESSIONAL
Device type	Fixed intrinsically safe HMI terminal	Fixed Windows Panel PC / Smart HMI
Display / screen	Small HMI display	24" Panel PC
Operating system	HMI terminal OS	Windows 11 IoT
Gas hazardous area category	II 1G	II 2G
Typical gas zone	Zone 0 / 1 / 2	Zone 1
Gas group	IIC	IIC
Temperature class	T4	T4
ATEX / IECEx gas marking	II 1G Ex ia IIC T4	II 2G Ex eb q [ib] IIC T4 Gb
Dust hazardous area	II 2D	II 2D Ex tb IIIC
Dust zone	Likely Zone 21 / 22 depending certificate	Zone 21 / 22
Communication	Terminal communication / Modbus depending model	Ethernet / USB / serial depending configuration

Camera Specification

Required
Zone 2, Group IIB,
T5 (100°C)



Existing CCTV



Item	MAXIMUS MVX DELUX	AXIS XFQ1656	Pelco ExSite Pro Bullet	Dahua IPC-HFE8849F-Z-PV
Resolution / image sharpness	2 MP / Full HD, 1920 × 1080.	4 MP, up to 2688 × 1512.	8 MP / 4K-class.	8 MP / 4K, 3840 × 2160
Sensor size	1/2.8" CMOS.	1/1.8" CMOS.	Not clearly stated in the source I used.	1/1.8" CMOS
Frame rate	Up to 60 fps.	Up to 50/60 fps.	up to 25/30 fps	Up to 25/30 fps at 8 MP
Zoom capability	30x optical zoom, 4.5–135 mm.	Varifocal lens 3.9–10 mm; not designed for long-range 30x zoom.	Bullet-style camera; the main strength is not long optical zoom.	Motorized varifocal 2.7–12 mm
Night / low-light performance	Strongest on paper for very dark scenes: 0.006 lux color / 0.0006 lux B/W.	Good low-light capability with Lightfinder and day/night function.	Has built-in IR , which helps in dark areas.	Very good low-light: 0.003 lux color / 0.0003 lux B/W; built-in IR + warm light up to 50 m
Backlight / high-contrast scenes	120 dB WDR.	Forensic WDR.	Very strong WDR, around 130–144 dB depending on mode/version.	120 dB WDR
Video compression	H.264 / MPEG4 / MJPEG / JPEG.	H.264 / H.265 with Axis Zipstream.	H.264 / H.265.	H.265, H.264, H.264H, H.264B, MJPEG; Smart H.265+/H.264+; AI H.265/H.264
AI / analytics readiness	Mainly a rugged surveillance camera ; not positioned as AI-native in the datasheet.	Strongest AI platform: deep-learning- based analytics , metadata, AXIS Object Analytics.	Analytics-enabled, including object detection / classification features.	Strong AI-ready camera: Face Detection, IVS, People Counting, ANPR, PPE Detection, Heat Map, SMD 4.0 AcuPick, AI SSA
Hazardous-area suitability	Zone 1/2 and Zone 21/22.	Zone 1 / Division 1 / Zone 21.	Explosion-proof zone 2	Zone 1 / Zone 21
Main strength	Best for long-distance zoom and very low-light surveillance .	Best overall balance of image quality + AI readiness + hazardous certification .	Best for highest image detail , with IR and modern analytics support.	Best for 8 MP + AI analytics + active deterrence , with built-in speaker, dual mic, IR/warm light, PPE detection and ANPR
ATEX Marking	Ex db IIC T5/T6 Gb c	Ex db I Mb Ex db IIC T5 Gb	Ex II 2G Ex db IIB+H2 T5 Gb	Ex II 2 G Ex db IIC T6 Gb Ex II 2 D Ex tb IIIC T80°C Db.

Site survey at gantry #15 & Server readiness

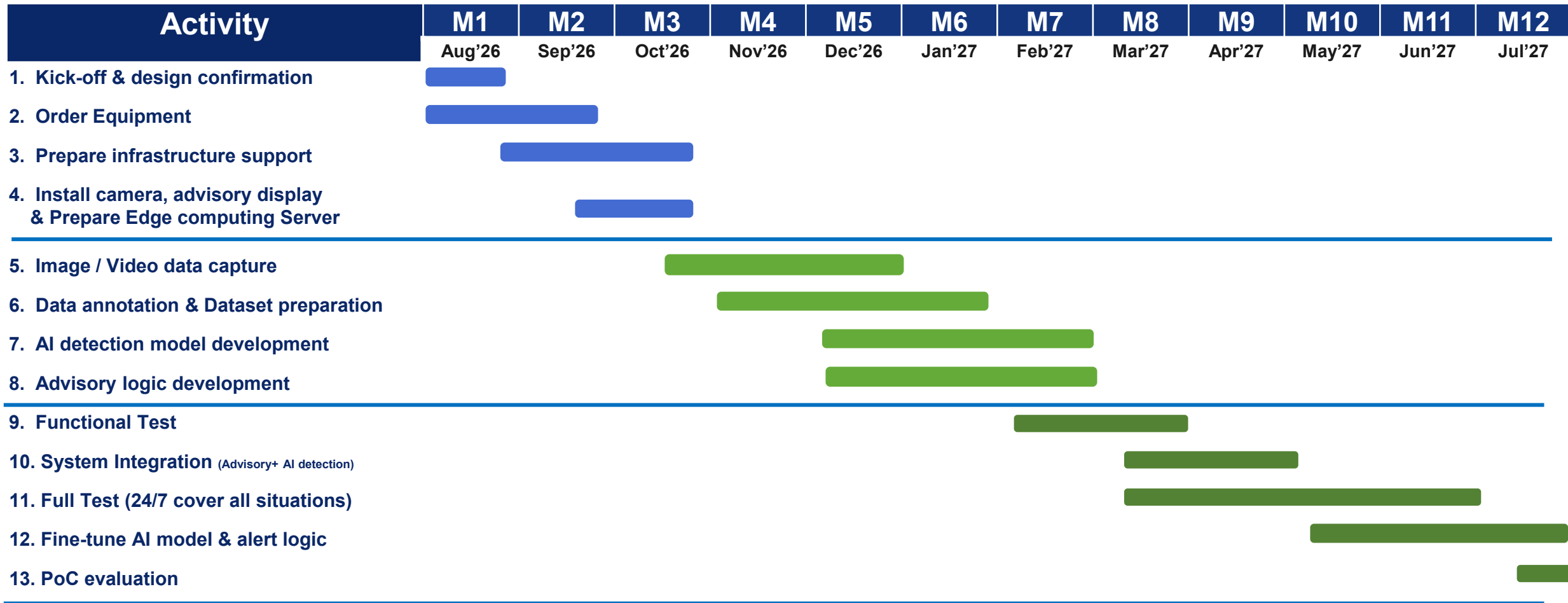
Existing ENIM Server :

1. Verify CPU / GPU / RAM / storage requirements
- 2.

LAN & Communication

EE

Project Milestones:



Key Deliverables



- PoC-ready **AI detection model** for loading arm and compartment
- **Advisory display logic** for recommended loading sequence
- Driver **compliance monitoring** result